

Tom Yujun Wu

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Education

Ph.D	University of North Carolina at Chapel Hill , Physics	Chapel Hill, US
	• Advisor: Prof. Carl Rodriguez • Research Interest: Gravitational Waves, Compact Objects, Astrodynamics	Aug 2025 -
Hons. B.Sc	University of Toronto , Astrophysics and Computer Science	Toronto, Canada
	• Advisor: Prof. Maya Fishbach • Degree: Astronomy and Physics Specialist, Computer Science Major • Graduated with High Distinction	Sep 2021 - May 2025
High School	High School Affiliated to Renmin University of China (RDFZ)	Beijing, China
	• Top High School in China	Sep 2018 - Jul 2021

First Author Publications

Are Long Gamma-Ray Bursts Progenitors to Merging Binary Black Holes?	ApJ
T. Y. Wu, M. Fishbach	2024
10.3847/1538-4357/ad98ed 🔗	

Co-authored and Collaboration Publications

Implications of low neutron star merger rates for gamma-ray bursts, r-process production and Galactic double neutron stars	ApJL accepted
M.Fishbach, A.P.Ji, W-F.Fong, T. Y. Wu , J.C.Rastinejad, A.Vijaykumar, H-Y.Chen	2026
10.48550/arXiv.2604.05059 🔗	
GWTC-4.0: Population Properties of Merging Compact Binaries	
<i>The LIGO Scientific Collaboration et al.</i> (including T. Y. Wu)	2025
10.48550/arXiv.2508.18083 🔗	
Contribution: calculated binary neutron star merger rates.	
News Coverage by CITA 🔗	
Upper Limits on the Isotropic Gravitational-Wave Background from the first part of LIGO, Virgo, and KAGRA's fourth Observing Run	2025
<i>The LIGO Scientific Collaboration et al.</i> (including T. Y. Wu)	
10.48550/arXiv.2508.20721 🔗	
Contribution: computed binary neutron star merger rates under specified spin-distribution assumptions.	

Contributed Talks

16th International LISA Symposium , MD, US	2026
• Abstract accepted for a contributed talk	

Research Experience

Graduate Student Researcher, University of North Carolina at Chapel Hill	Chapel Hill, US
• Advisor: Prof. Carl Rodriguez	Aug 2025 -

- Using Cluster Monte Carlo (CMC) and COSMIC to study the impact of intermediate-mass black holes (IMBHs) in globular clusters.
- Investigating how IMBHs affect compact-object populations, binary black hole mergers, and extreme mass-ratio inspirals (EMRIs) relevant to LISA.

Undergraduate Researcher, University of Toronto

Toronto, Canada
Aug 2023 – Jun 2025

- Advisor: Prof. Maya Fishbach
- **Long gamma-ray bursts and binary black holes:** Estimated merger delay-time distributions, formation efficiencies, and spin-dependent effects.
- **Binary neutron star merger rates:** Analyzed merger-rate constraints under different astrophysical models and assumptions, using information from transients and Galactic binary neutron stars.
- Resulted in one first-author publication and one additional manuscript in preparation.

Undergraduate Researcher (Summer), University of Toronto

Toronto, Canada
May 2024 – Sep 2024

- Advisors: Prof. Ziqing Hong and Prof. Miriam Diamond
- Simulated detector-electronics behavior for the SuperCDMS experiment using computing clusters.
- Developed a pipeline to compare simulations with acquired data and improve detector-response modeling.
- Presented this work at the Canadian Astroparticle Physics Summer Student Talk (CASST) at SNOLAB.

Summer Research Exchange Student, National University of Singapore

Singapore
May 2023 – Jul 2023

- Advisor: Prof. Alexey Berdyugin
- Studied the use of machine-learning techniques to identify and map two-dimensional materials such as graphene.
- Worked on software and hardware tools to automate the graphene flake-search process.

Awards and Scholarships

Travel Grant - 16th International LISA Symposium

2026

- \$1295 USD from NASA

University of Toronto - Dean's List Scholar

2022, 2023, 2024, 2025

Victoria College - The Regents In-course Scholarship

2022, 2023, 2024

Undergraduate Summer Research Exchange Award

2023

University of Toronto Admissions Scholarship

2021

The International Mathematical Modeling Challenge

2020

- Global Meritorious Award, Top 4 among 54 invited teams from 30 countries.
- Greater China Region Outstanding Award, Top 8 among 650 teams.